WEB DEVELOPMENT 1

Learning Groups: P107, P108, E101, P102

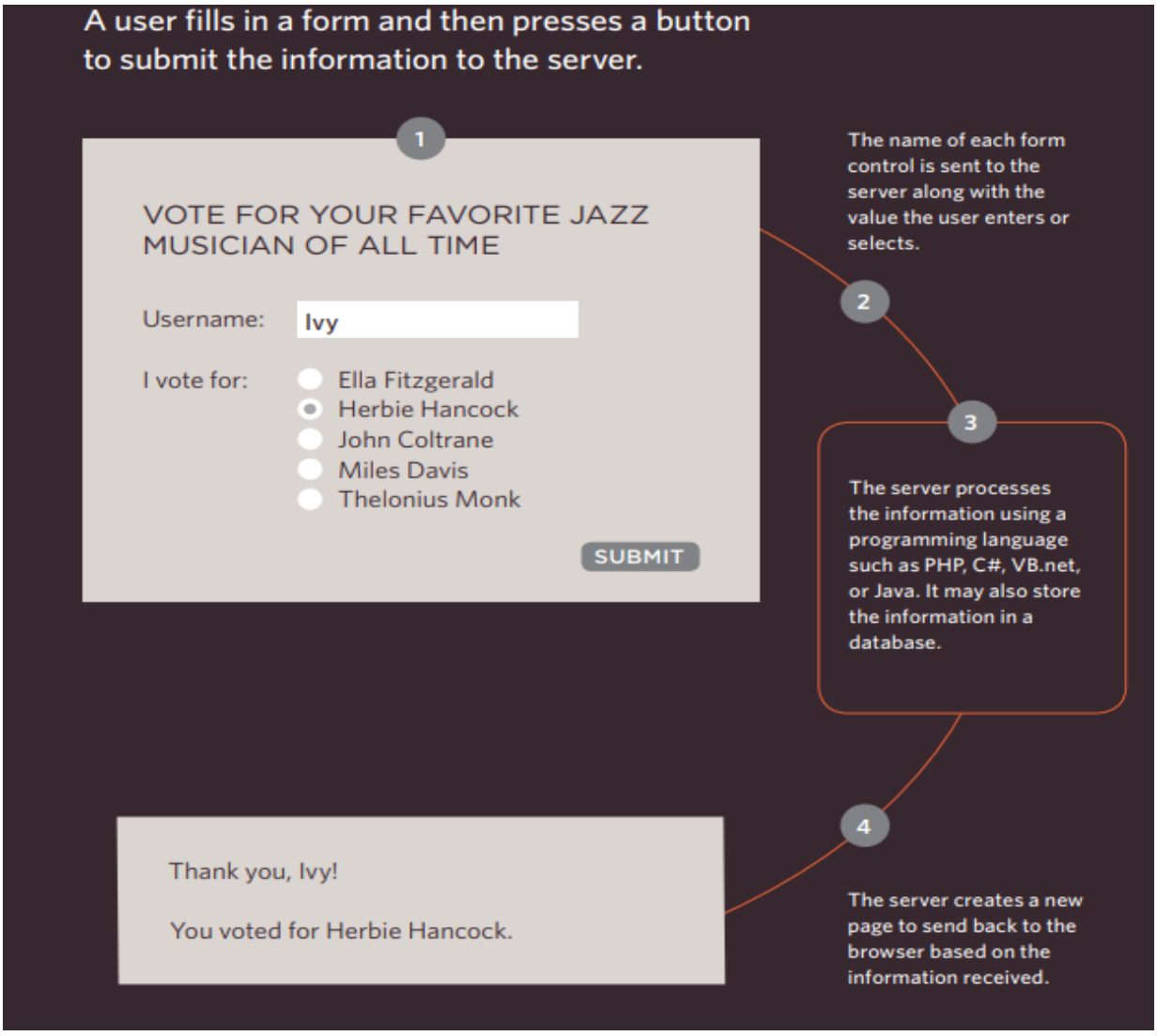
Lecturer: Mr. Maina Samuel

week 3 **learning topics**

- **4.0 HTML forms** **input attributes, form attributes.**
- **4.1 HTML frames**
- **4.2 HTML i frame (inline frames)**
-

4.0 HTML forms

HTML forms are used to collect user input.



The **<form>** element defines an HTML form:

```
<form>
.
form elements
.
</form>
```

Form elements- are different types of input elements, checkboxes, radio buttons, submit buttons, and more.

1. **The <input> Element**

The **<input>** element is the most important **form element**.

The <input> element has many variations, depending on the **type** attribute.

Input type attribute. Examples.

Text	Defines normal text input
Radio	Defines radio button input (for selecting one of many choices)
Submit	Defines a submit button (for submitting the form)
Password	
Checkbox	

Text Input

<input type="text"> defines a one-line input field for **text input**:

Edit This Code: See Result » Result:

```
<!DOCTYPE html>
<html>
<body>

<form>
  First name:<br>
  <input type="text" name="firstname">
  <br>
  Last name:<br>
  <input type="text" name="lastname">
</form>

<p>Note that the form itself is not visible.</p>

<p>Also note that the default width of a text input field is 20 characters.
</p>

</body>

</html>
```

First name:

Last name:

Note that the form itself is not visible.

Also note that the default width of a text input field is 20 characters.

2. **Radio Button Input**

<input type="radio"> defines a **radio button**.

Radio buttons let a user select ONE of a limited number of choices:

Edit This Code: See Result » Result:

```
<!DOCTYPE html>
<html>
<body>

<form>
  <input type="radio" name="gender" value="male" checked> Male<br>
  <input type="radio" name="gender" value="female"> Female<br>
  <input type="radio" name="gender" value="other"> Other
</form>

</body>

</html>
```

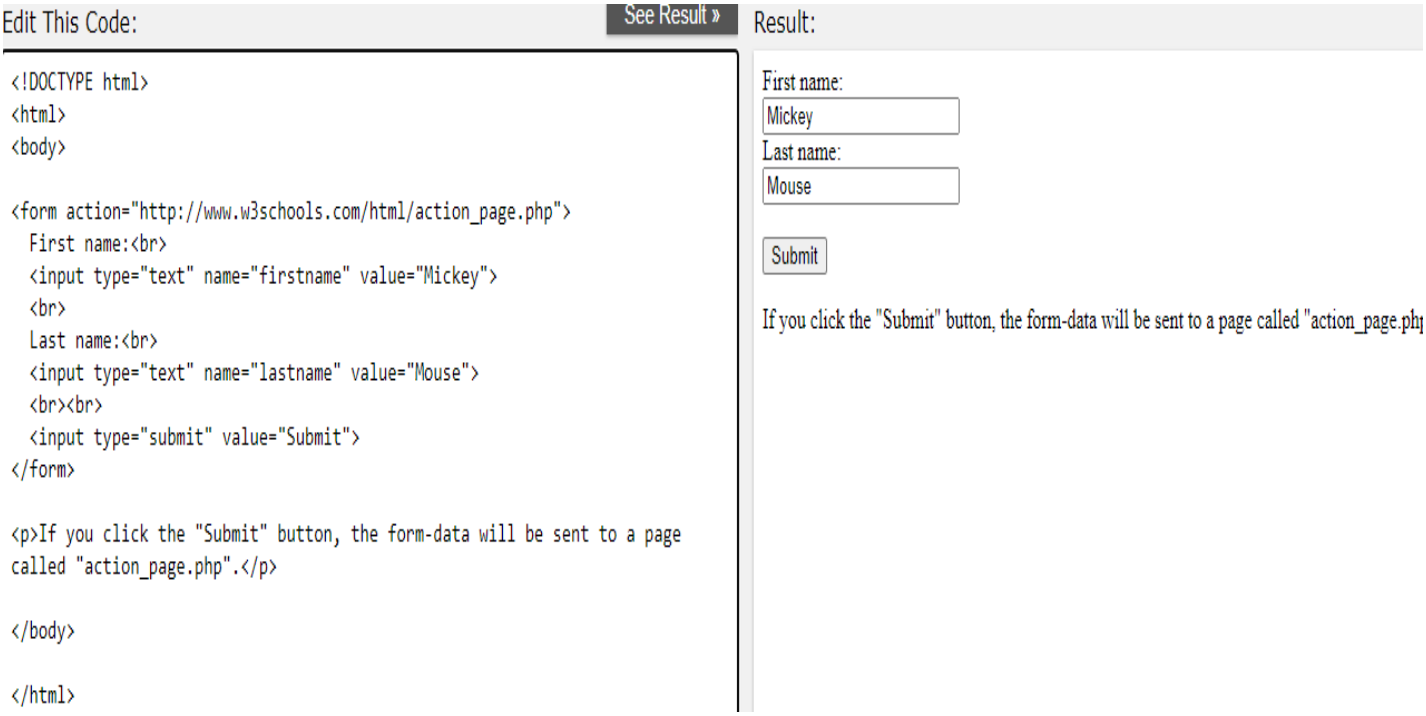
(●) Male
( ) Female
( ) Other

3. The Submit Button

<input type="submit"> defines a button for **submitting** a form to a **form-handler**.

The form-handler is typically a server page with a script for processing input data.

The form-handler is specified in the form's **action** attribute:



The Action Attribute

The **action attribute** it's used to specify the URL of the document where the data to be sent after the submission of the form. Can be used in the <form >element

<form action="**url** ">

```
<form action="action_page.php">
```

Possible values of the URL

- Absolute URL – An absolute URL contains the entire address from the protocol (HTTPS) to the domain name (www.example.com)
- Relative URL- A Relative File Path points to an asset, relative to the current page <img src="/img/photo.jpg" alt="Photo of Something">

When Should You Use Absolute or Relative file paths?

Let's break it down like this, if you are

- linking to content on your website, it's often best to use a Relative file path.
- If you're linking to an external source or website, then use an Absolute URL.

Why use absolute url?

1. **Makes it difficult to clone your site**

Cloned websites are used by cybercriminals to defraud a company's customers. To create clones, they use scraping tools that search and copy a site's content. Absolute URLs make it more difficult for them to do this.

2. **Stops duplicate content**

Duplicate website content can impact your SEO and your site's ranking. Absolute URLs help prevent duplicate content issues. Search engines, for example, consider the http://, http://www, https:// and https://www versions of a website as different websites. If all these versions were crawled and the site used relative URLs, it could lead the search engines to index four versions of each of your web pages. As absolute URLs contain the specified protocol, this cannot happen.

3. **Assists with search engine indexing**

Search engine crawlers index your site by following internal links. However, they don't always search the whole site. Using absolute URLs, however, makes it easier for them to index more of your site quickly so that more of your pages are searchable online.

Why use Relative url?

1. **Faster coding and development**

You can code a website far easier if you only need to use, shorter relative URLs. If you are developing a website and have live and developmental versions with different domains, relative URLs allow you to bring your newly developed site online without having to modify all the URLs to those of your live site.

2. **Internal linking**

Remember, relative URLs can only be used to link to resources with the same root directory

Form Attributes

ATTRIBUTE	VALUE
name	This attribute specifies the name of the form.
action	specifies the destination URL
method	This attribute specifies method by which data will be passed to server. Example -GET and POST
enctype	This attribute species encryption type for the data before it is passed to the server.

HTTP Request Methods

The Hypertext Transfer Protocol (HTTP) is designed to enable communications between clients and servers.

HTTP works as a request-response protocol between a client and server.

Example: A client (browser) sends an HTTP request to the server; then the server returns a response to the client. The response contains status information about the request and may also contain the requested content.

HTTP Methods

- **GET**
- **POST**
- **PUT**
- **HEAD**
- **DELETE**
- **PATCH**
- **OPTIONS**

The two most common HTTP methods are: GET and POST.

The Method Attribute

The **method attribute** specifies the HTTP protocol method (**GET** or **POST**) to be used when submitting the forms:

<form method="get" action="page.html"> </form>

Or

<form method="post" action="page.html"> </form>

The GET Method

GET is used to request data from a specified resource.

GET is one of the most common HTTP methods.

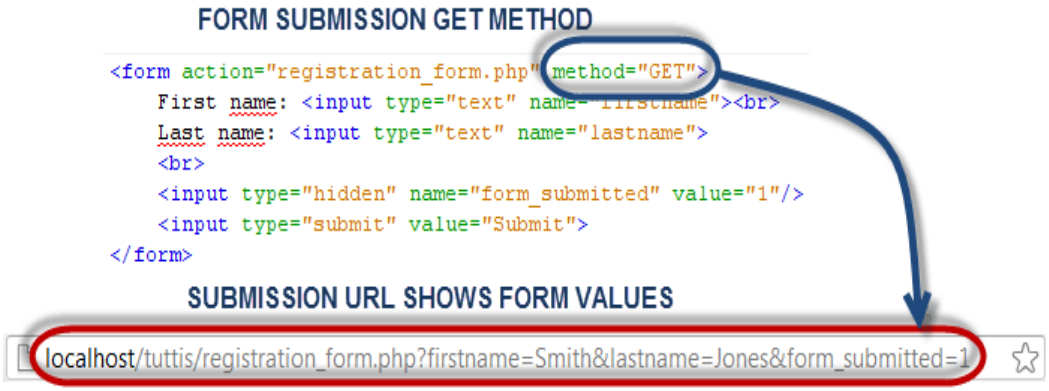
Note that the query string (name/value pairs) is sent in the URL of a GET request: https:

```
/test/demo_form.php? name1=value1&name2=value2

//www.xul.fr/page.html? color=blue shape=rectangle
```

Some other notes on GET requests:

- GET requests can be cached
- GET requests remain in the browser history
- GET requests can be bookmarked
- GET requests should never be used when dealing with sensitive data
- GET requests have length restrictions
- GET requests are only used to request data (not modify)



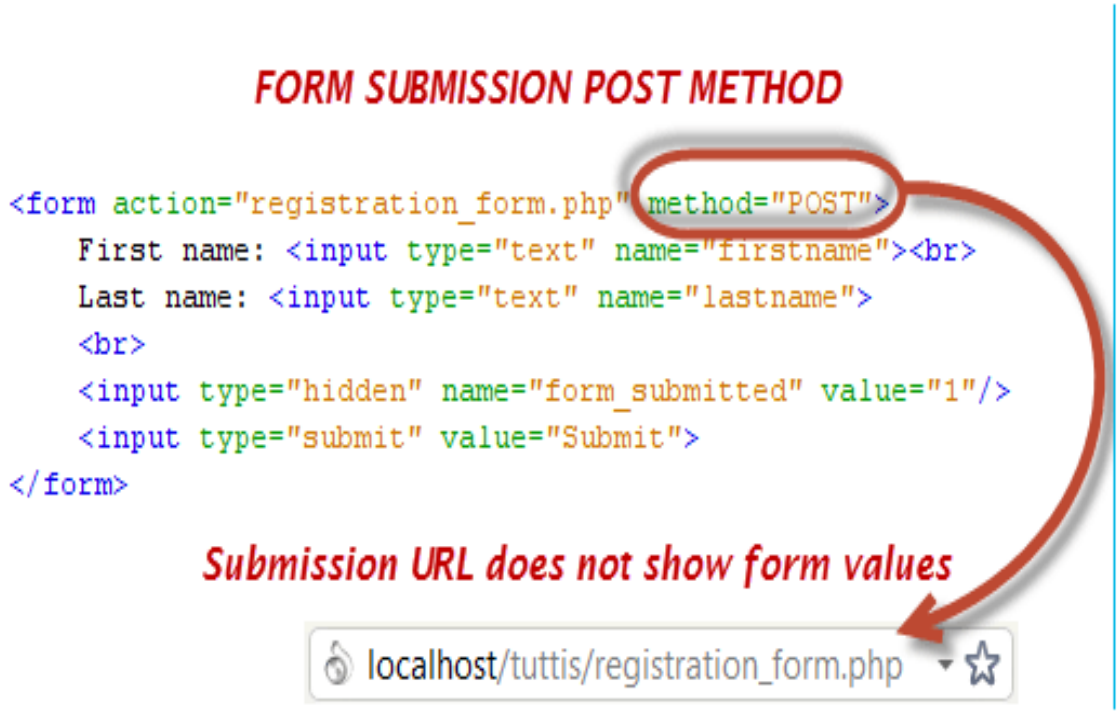
The POST Method

POST is used to send data to a server to create/update a resource.

The data sent to the server with POST is stored in the request body of the HTTP request:

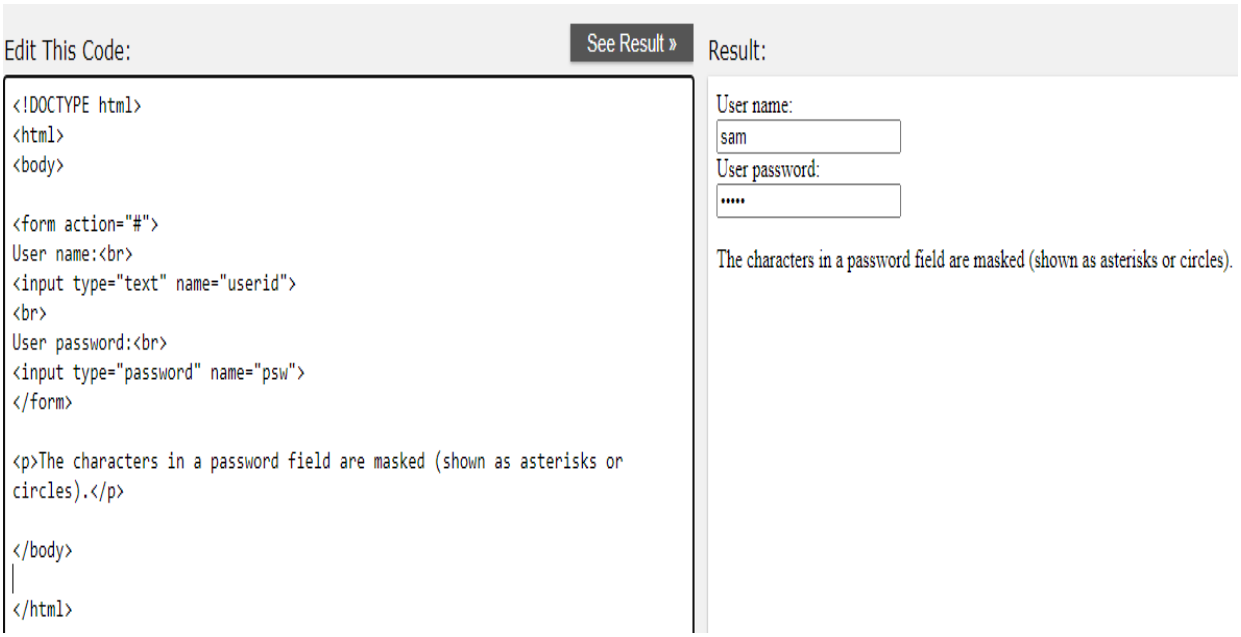
Some other notes on POST requests:

- POST requests are never cached
- POST requests do not remain in the browser history
- POST requests cannot be bookmarked
- POST requests have no restrictions on data length



Input Type: password

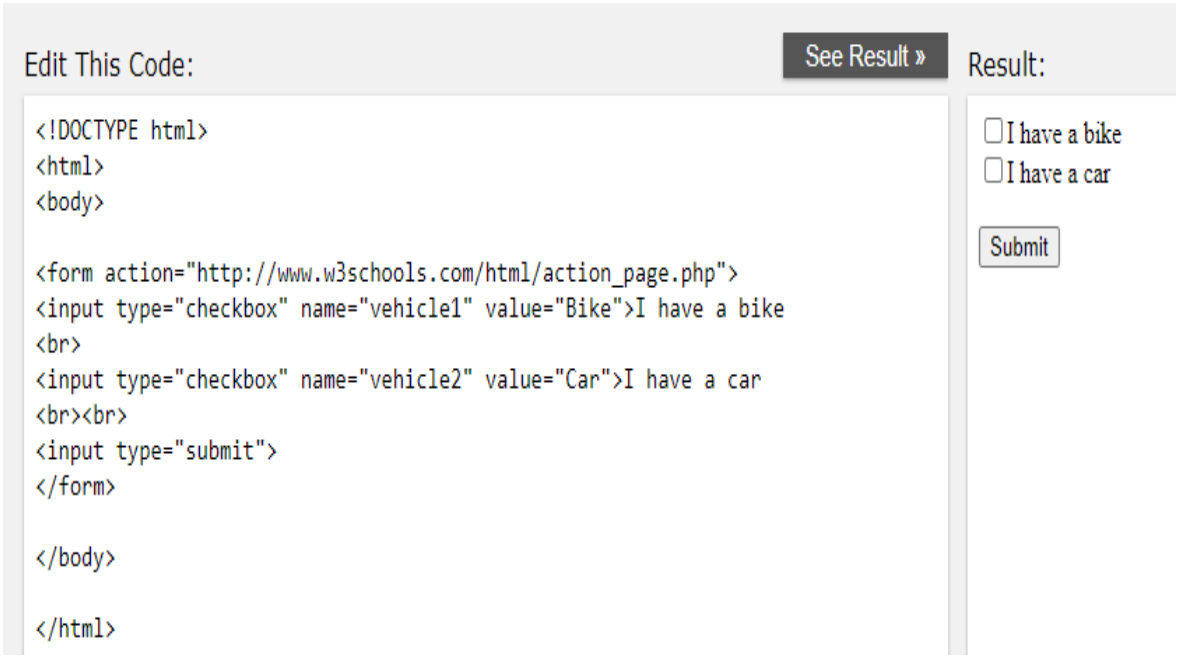
<input type="password"> defines a **password field**:



Input Type: checkbox

<input type="checkbox"> defines a **checkbox**.

Checkboxes let a user select ZERO or MORE options of a limited number of choices.



Example 1: A simple form to demonstrate the concept learned

Edit This Code: See Result » Result:

```
<html>
    <head>
        <title>
            Example of Form tag
        </title>
    </head>
    <body>
        <h2>Simple Form</h2>
        <form>
            <div>Enter your Name:</div>
            <div><input type="text" name="name"/></div>
            <div>Share something about yourself:</div>
            <div><textarea name="bio"></textarea></div>
            <div>
                I like:
                <select>
                    <option>Football</option>
                    <option>Cricket</option>
                    <option>Hockey</option>
                </select>
            </div>
            <div><input type="submit" value="Submit"/></div>
        </form>
    </body>
</html>.
```

Simple Form

Enter your Name:

Share something about yourself:

I like: Football

Subm

Football
Cricket
Hockey

Example 2: creating a customer Review form

form.html

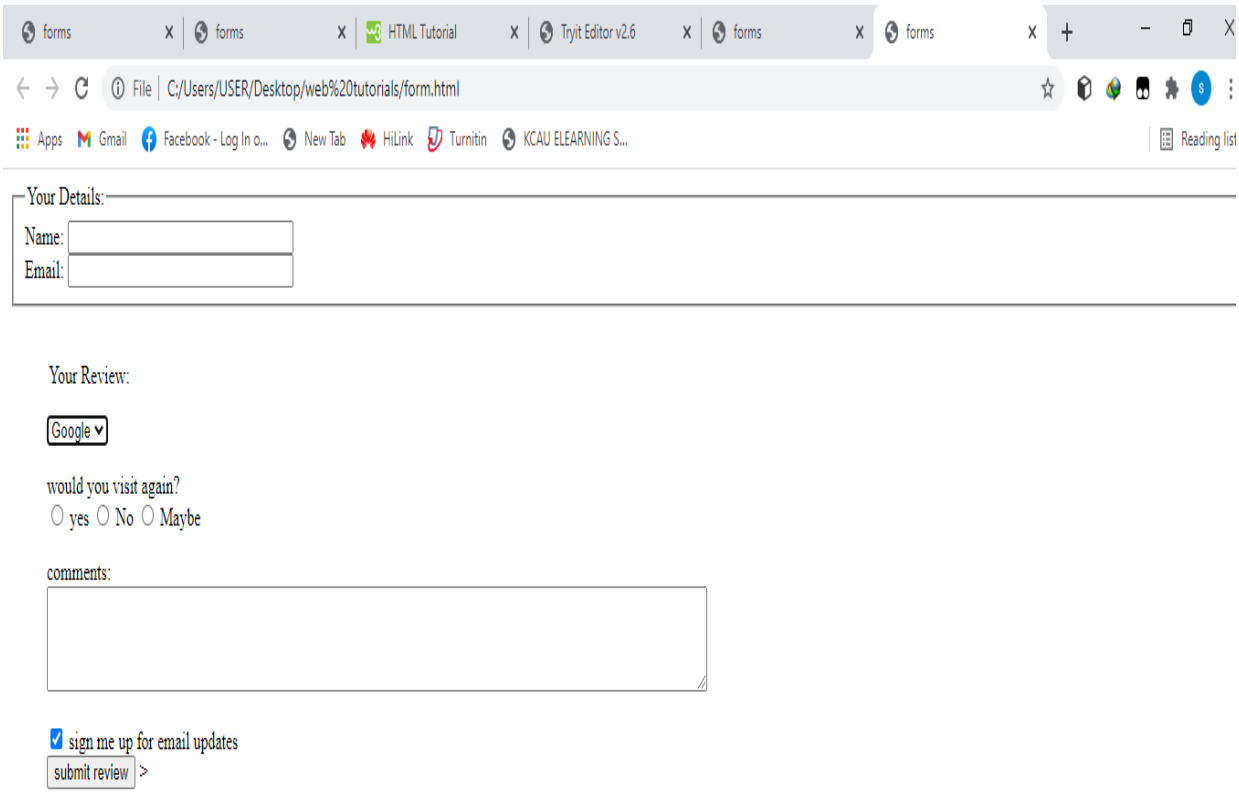
```
<!DOCTYPE html>
<html>
<head>
    <title>forms</title>
</head>
<body>
    <form action="http"://www.example.com/review.php"method="get">
        <fieldset>
            <legend>
                Your Details:
            </legend>
            <label>
                Name:
                <input type="text" name="name"size="30"maxlength="100">
            </label>
            <br/>
            <label>
                Email:
                <input type="email" name="email"size="30"maxlength="100">
            </label>
            <br/>
        </fieldset>
        <br/>
        <figure>
            <legend>
                Your Review:
            </legend>
            <p>
                <label for="hear-about">
                </label>
```

```
                <select name="referrer" id="hear-about">
                    <option value="google">Google</option>
                    <option value="Friend">Friend</option>
                    <option value="advert">Advert</option>
                    <option value="other">Other</option>
                </select>
            </p>
            <p> would you visit again?
                <br/>
                <label>
                    <input type="radio" name="rating"value="yes"/>
                    yes
                </label>
                <label>
                    <input type="radio" name="rating"value="no"/>
                    No
                </label>
                <label>
                    <input type="radio" name="rating"value="maybe"/>
                    Maybe
                </label>
            </p>
            <p>
                <label for ="comments">
                    comments:
                </label>
                <br/>
                <textarea rows="4 "cols="40 "id="comments">
                </textarea>
            </p>
```

```
            <label>
                <input type="checkbox" name="subscribe"checked="checked"/>
                sign me up for email updates
            </label>
            <br/>
            <input type="submit" value="submit review"/>
            </fieldset>
        </form>


</body>
</html>>
```

Output



4.1 HTML frames

HTML frames are used to divide your browser window into multiple sections where each section can load a separate HTML document. Or a sub window inside an active browser. possible to have more than two frames in a web page.

A collection of frames in the browser window is known as a **frameset**. The window is divided into frames in a similar way the tables are organized: into rows and columns.

Frames were introduced in order to aid navigation through complex webpages. For example, one frame could hold a menu of links, while their associated web pages are displayed in another frame.

Disadvantages of Frames

There are few drawbacks with using frames, so it's never recommended to use frames in your webpages −

- Some smaller devices cannot cope with frames often because their screen is not big enough to be divided up.

- Sometimes your page will be displayed differently on different computers due to different screen resolution.
- There are still few browsers that do not support frame technology.

Note: Do not use HTML <frame> tag as it is not supported in HTML5, instead you can use <iframe> or with CSS to achieve similar effects in HTML.

Syntax **HTML <frameset>** **<frameset cols=" ">.....</frameset>**

< frame src = "URL" > ***hint frame has no <body>tag**

Creating vertical frames. **Create web folder and using sublime txt save all frame documents in one folder.**

1. Step **one** index frame page that will be used to run the linked src frames as frame.html

```
form.html                                    frame.html
<!DOCTYPE html>
<html>
<head>
    <title>Frame tag</title>
</head>
  <frameset cols="25%,50%,25%">
    <frame src="frame1.html" >
    <frame src="frame2.html">
    <frame src="frame3.html">
  </frameset>
</html>
```

2. Step **two** save as frame1.html

```
form.html    frame.html    frame2.html
<!DOCTYPE html>
<html>
<head>
    <style>
       div{
                                background-color: #7fffd4;
         height: 500px;
        }
    </style>
</head>
<body>
    <div>
        <h2>This is first frame</h2>
    </div>
 </body>
</html>
```

3. Step three save as frame2.html

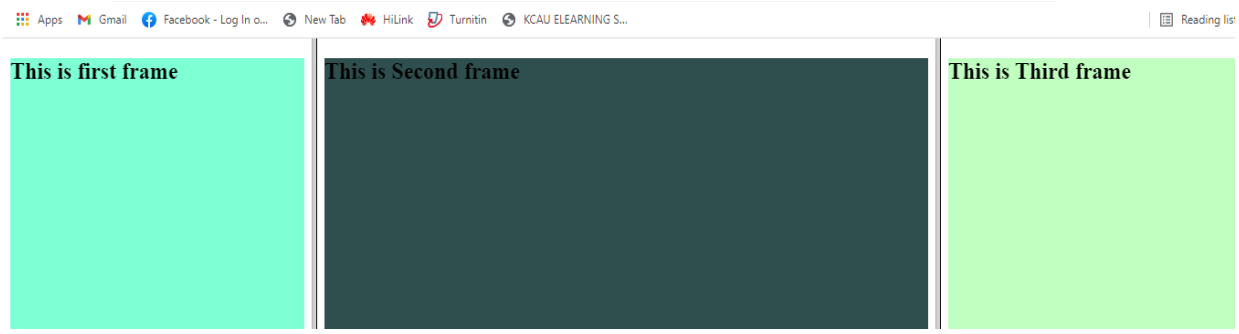
```
<!DOCTYPE html>
<html>
<head>
    <style>
        div{
            background-color: #2f4f4f;
            height: 500px;

        }
    </style>
</head>
<body>
    <div>
        <h2>This is Second frame</h2>
    </div>
 </body>
</html>
```

4. Step four save as frame3.html

```
<!DOCTYPE html>
<html>
<head>
    <style>
        div{
            background-color: #c1ffc1;
            height: 500px;
                    }
    </style>
</head>
<body>
                <div>
        <h2>This is Third frame</h2>
    </div>
 </body>
</html>
```

Output



4.2 HTML i frame (inline frames)

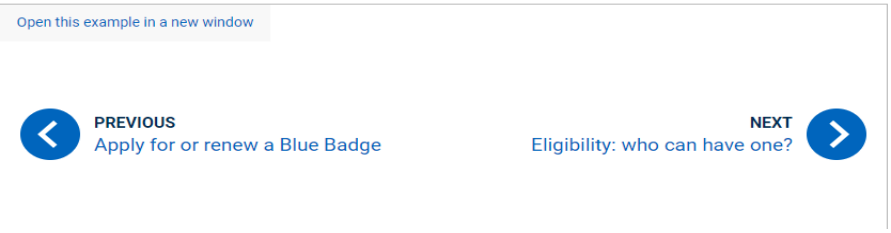
What is i frame?

To put it simply, it's a separate web page inside of your existing web page.
Examples of iframes include tweets embedded into news articles, YouTube videos embedded into a blog post, sidebar ads, social media buttons – the list is nearly endless.

Why are frames inHTML5 OBSOLETE? (entirely removed)

Framesets are often used in a way where they break the fundamental principle of the Web - that each document has a **single URL** while current screen is now determined by a *sequence* of navigation actions rather than a *single* navigation action. This leads to problems with linking, bookmarks, search engines.

What is sequence navigation A 'next' and 'previous' link to pages that come before or after the user's current web page...*example of sequence navigation*



What's the difference between frame and i frame?

Frame	Iframe
Frame is part of a **Frameset**.	is the fact that an **Iframe** is an inline **frame**
Frames are a bunch of 'boxes' put together to make one site with many pages	Inline **frame** is just one "box" and you can place it anywhere on your site.
frame cannot be placed in the body	***iframe*** tag is that it can be used to load external html or xhtml pages into the current document without disturbing the original page.

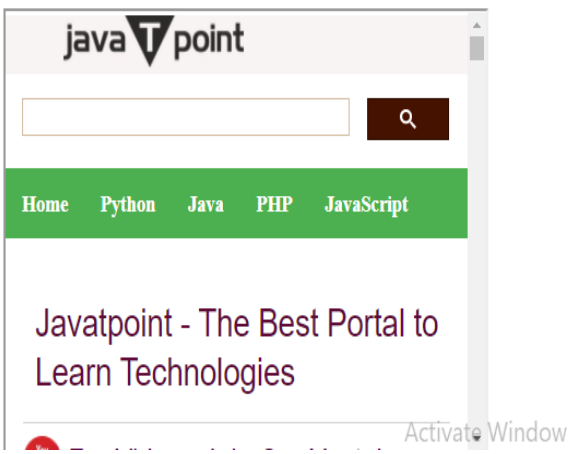
Uses of frames

- To embed and isolate third-party content into a website.
- Most of web advertising solutions are based on **iframes** - because they give security (cross-domain policy) and isolated rectangle on screen which can be fully managed by third party content and scripting (a common use case is advertisements).

Example 1 illustrating iframe with dimensions' width and height.

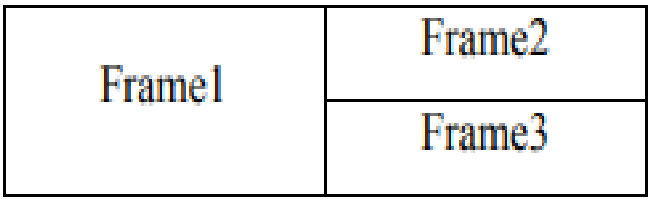
Run ▶ Online Code Editor

```
<!DOCTYPE html>
<html>
<body>
<h2>HTML Iframes example</h2>
<p>Use the height and width attributes to specify the size of the
iframe:</p>
<iframe src="https://www.javatpoint.com/" height="300" width="400">
</iframe>
</body>
</html>
```



Revision Questions

1. Create a web page which should generate following output:



2. . Create a simple form

    - accepting Name
    - Enrollment No.
    - and Submit button

NB: these individual's practical revision exercises if tried will improve your coding skills.

Next topic lesson will be on java script since CSS (cascading style sheet) have been tacked inside html lessons then week 7 online CAT but you can give your views I receive them from your class reps.